

Telecom Churn Analysis

1. Basic Data Understanding

To begin any churn analysis, it is crucial to understand the overall churn percentage within the customer base.

- First, identify all customers whose churn status is marked as "Yes" and count them.
- Then, divide this number by the total number of customers to calculate the overall churn rate.
- This metric provides a foundation for understanding the scale of the churn problem and sets the baseline for deeper analysis.

Churn_rate : 26.5784982935153584

Step 2: Cohort Analysis (Tenure-Based)

Cohort analysis helps reveal when customers are most likely to leave.

- Create tenure-based cohorts (e.g., 0–6 months, 7–12 months, 1–2 years, etc.) to group customers with similar service durations.
- Compare churned customers vs. total customers within each tenure segment.

- If early-tenure cohorts show significantly higher churn, it indicates issues in onboarding, expectation-setting, service quality, or pricing clarity early in the customer lifecycle.
- This helps the company prioritize improvements during the initial months of service.

1	tenure_group	total_churn	churned_customer	churn_rate
2	0-12 month	2175	1037	47.68
3	13-24 month	1024	294	28.71
4	25-48 month	1594	325	20.39
5	49+month	2239	213	9.51

Step 3: Customer Segmentation (Risk-Based)

Segmenting customers based on behavioral and demographic patterns helps identify who is most likely to churn.

- Customers on month-to-month contracts, paying higher monthly charges, and having short tenure typically fall into the high-risk category.

- Customers with long-term contracts, stable tenure, and consistent usage patterns can be classified as loyal or low-risk.
- Medium-risk customers fall between these extremes and often respond well to targeted interventions.
- This segmentation enables more personalized strategies, such as tailored discount offers, early engagement programs, or proactive retention calls.

1	customerid	Contract	MonthlyCharges	tenure	Churn	risk_segment
2	7590-VHVEG	Month-to-month	29.85	1	No	medium risk
3	5575-GNVDE	One year	56.95	34	No	stable
4	3668-QPYBK	Month-to-month	53.85	2	Yes	churned
5	7795-CFOCW	One year	42.3	45	No	stable
6	9237-HQITU	Month-to-month	70.7	2	Yes	churned
7	9305-CDSKC	Month-to-month	99.65	8	Yes	churned
8	1452-KIOVK	Month-to-month	89.1	22	No	medium risk
9	6713-OKOMC	Month-to-month	29.75	10	No	medium risk
10	7892-POOKP	Month-to-month	104.8	28	Yes	churned
11	6388-TABGU	One year	56.15	62	No	stable
12	9763-GRSKD	Month-to-month	49.95	13	No	medium risk
13	7469-LKBCI	Two year	18.95	16	No	stable

:- More Data is there its only for showcase

Step 4: Segment Churn Rates by Contract Type

Understanding how contract types influence churn can guide retention strategies.

- Month-to-month customers typically reflect the highest churn rate because they have greater flexibility to switch providers.
- In contrast, one-year or two-year contract customers usually exhibit far lower churn due to contractual commitments.
- Insights from this step can fuel targeted retention campaigns—like offering discounts, loyalty points, or upgrades to encourage long-term contract adoption.

1	Contract	total_customers	churned_customer	churn_rate
2	One year	1472	166	11.28
3	Month-to-month	3875	1655	42.71
4	Two year	1685	48	2.85

Step 5: Payment Method and Churn Behavior

Payment methods can strongly correlate with churn patterns.

- Customers using electronic checks often show higher churn rates compared to those using credit cards or auto-pay options.
- This segment might require special attention—for example, encouraging them to switch to auto-pay, which reduces missed payments and increases customer stickiness.
- Understanding payment preferences helps minimize churn caused by payment-related issues.

1	PaymentMethod	total_customers	churned_customer	churn_rate
2	Electronic check	2365	1071	45.29
3	Credit card (automatic)	1521	232	15.25
4	Mailed check	1604	308	19.20
5	Bank transfer (automatic)	1542	258	16.73